FLATNESS OF A MODULE ON A RINGED SPACE IS DETECTED ON THE STALKS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The description of the stalk of a tensor product of modules as the tensor product of the stalks. Invoked twice in the counted proof (source lines 2781-2783 and 2791-2792) as a pointer to a lemma of this chapter. Transfers exactness of a sequence of modules to exactness after tensoring, stalk by stalk.
- The skyscraper sheaf at a point associated to a module over the local ring. Used inside the counted proof (source lines 2784-2795). Realises an arbitrary module over the local ring as the stalk of a sheaf, which tests flatness of the stalk.

Source reference. Original lemma lemma-stalk-tensor-product: sources/stacks-modules/pinned-original-source.tex, source lines 2604–2604. Navigation pointer only: the counted text refers to the Stacks item lemma-stalk-tensor-product, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. An $\mathcal{O}_X$-module $\mathcal{F}$ is flat if and only if the stalk $\mathcal{F}_x$ is a flat $\mathcal{O}_{X,x}$-module for all $x \in X$.*

Proof. Assume $\mathcal{F}_x$ is a flat $\mathcal{O}_{X,x}$-module for all $x \in X$. In this case, if $\mathcal{G} \to \mathcal{H} \to \mathcal{K}$ is exact, then also $\mathcal{G} \otimes_{\mathcal{O}_X} \mathcal{F} \to \mathcal{H} \otimes_{\mathcal{O}_X} \mathcal{F} \to \mathcal{K} \otimes_{\mathcal{O}_X} \mathcal{F}$ is exact because we can check exactness at stalks and because tensor product commutes with taking stalks, see Lemma lemma-stalk-tensor-product. Conversely, suppose that $\mathcal{F}$ is flat, and let

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$x \in X$. Consider the skyscraper sheaves $i_{x,*}M$ where M is a $\mathcal{O}_{X,x}$-module. Note that

$$M \otimes_{\mathcal{O}_{X,x}} \mathcal{F}_x = (i_{x,*}M \otimes_{\mathcal{O}_X} \mathcal{F})_x$$

again by Lemma lemma-stalk-tensor-product. Since $i_{x,*}$ is exact, we see that the fact that $\mathcal{F}$ is flat implies that $M \mapsto M \otimes_{\mathcal{O}_{X,x}} \mathcal{F}_x$ is exact. Hence $\mathcal{F}_x$ is a flat $\mathcal{O}_{X,x}$-module. $\square$